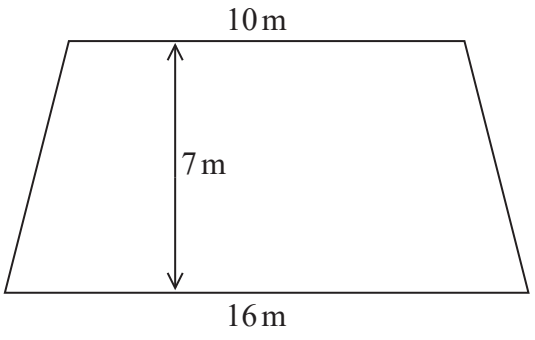
24 The diagram shows a floor in the shape of a trapezium.





£16.99

John is going to paint the floor.

Each 5 litre tin of paint costs £16.99
1 litre of paint covers an area of 2 m^2

John has £160 to spend on paint.

Has John got enough money to buy all the paint he needs?
You must show how you get your answer.

Trapezium area = ½(a + b)h

= ½ × (10 + 16) × 7 = 91 m^2

1 mark
Use the trapezium formula

	Litres	Area m^2	
×45.5	1	2	×45.5
	45.5	91	

1 mark
Convert area to litres

	Litres	Tins	
×9.1	5	1	×9.1
	45.5	9.1	

1 mark
Convert litres to tins

9.1 tins → 10 whole tins

	Tins	GBP	
×10	1	16.99	×10
	10	169.90	

1 mark
Round up, then cost the whole tins

No. £169.90 > £160, so he is £9.90 short.

1 mark
Correct cost and clear decision

(Total for Question 24 is 5 marks)